

Computer Networks (CS3001)

Final Exam Solution

Question # 1:

[40 Minutes]

[20 x 0.5 = 10 Marks]

- i. True - Packet switching allows dynamic sharing of bandwidth, which is ideal for bursty data. It doesn't require dedicated circuits and thus is more efficient for data like web browsing or email.
- ii. False - The application layer provides services directly to users. The transport layer provides communication services between processes on end systems, not directly to users.
- iii. False - DNS uses a distributed and hierarchical database system, not a centralized one, to ensure scalability and resilience.
- iv. True - P2P systems allow peers to request and serve content, contributing to scalability.
- v. False - The congestion window grows exponentially during slow start.
- vi. False - The data plane determines local forwarding; the control plane sets up the logic and forwarding tables.
- vii. True - These are examples of middleboxes performing non-standard routing functions.
- viii. True - DASH divides video into segments and lets the client select the best quality segment based on current bandwidth.
- ix. False - OSPF is an intra-AS routing protocol; BGP is used for inter-AS routing.
- x. True - In SDN, a remote controller installs forwarding rules in routers, separating control from forwarding.
- xi. False - MAC addresses are used locally within the same link; IP addresses are used across the Internet.
- xii. False - Third-party cookies can track users across multiple websites if those sites include content from the same tracker.
- xiii. True - The HEAD request fetches headers without the body, often used to check for modifications.
- xiv. False - A CNAME maps an alias name to a canonical name, not directly to an IP address.
- xv. False - BGP is a path-vector protocol used between autonomous systems, not within.
- xvi. False - BGP is a path-vector protocol, not hop-count based like RIP. It prevents loops by maintaining path information, not by limiting hop counts.
- xvii. False - OSPF selects the least-cost path, not based on congestion but administrative weights (static costs) assigned to links.
- xviii. True - MAC addresses are hard-coded into network interfaces by manufacturers and are independent of network location.
- xix. False - Ethernet is a connectionless protocol. It does not establish a connection before sending data.
- xx. False - Switches forward broadcast frames only when necessary (e.g., ARP), and protocols like Spanning Tree Protocol (STP) prevent broadcast storms.

Question # 2:

[40 Minutes]

[5 x 2 = 10 Marks]

Part (a) Solution:

Traffic intensity (I) = $(aL)/R = (30 \times 8 \times 8000)/(100,000,000) = 0.0192$.

Traffic intensity < 1 , system is under loaded $\rightarrow$ low queueing.

Part (b)

- i. Answer: A third-party persistent cookie.
- ii. Answer: Because both nytimes.com and socks.com embed content from the same third-party domain adx.com, the browser sends the cookie ID=7493 during both visits. This allows adx.com to track Susan across unrelated websites.

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Part (c)

- Answer: The CNAME allows netcinema.com to outsource content delivery to a third-party CDN (KingCDN.com) without exposing internal routing logic or maintaining their own servers. Bob ultimately downloads the video from a KingCDN edge server, not from netcinema.com directly.
- Answer: This mechanism provides load balancing, geographic optimization, and performance scaling. By redirecting Bob to KingCDN.com, the video can be fetched from a server closer to Bob, reducing latency and network congestion.

Part (d) Solution:

RR Inserted in Authoritative NS

· (StudyZone.org, 192.168.100.20, A)

· (StudyZone.org, 192.168.100.30, MX)

RR Inserted in TLD

· (StudyZone.org, dns1.bluehost.com, NS)

· (dns1.bluehost.com, 203.0.113.10, A)

Part (e) Answer:

The packet size is $L=10KB=10*8*Kbits=80 Kbits$

The bandwidth is $R=1Mbps=1000 kbps$

The transmission time is calculated as:

$L/R=80 Kb/1000 kbps =0.08 sec=80 ms$

Thus, the Utilization is determined as follows:

$U = (L/R)/(RTT+L/R) = 80/80+250=0.24$

Question # 3:

[70 Minutes]

[5 x 4 = 20 Marks]

Part (a)

- Solution: Part D. It should be the part of the code that a fresh ACK is received.
- Solution: Part C. It should be the part of the code that a timeout event occurs.
- Solution: Part E. It should be the part of the code that 3 duplicate ACKs are detected.
- Solution: Part C and E. It should be the part of the code that a packet loss is detected, whether it is due to timeout or 3 duplicate ACKs.

Part (b)

- Answer:
Slow Start: TCP slow start is operating in the intervals [1,6] and [23,26].
Congestion Avoidance: TCP congestion avoidance is operating in the intervals [6,16] and [17,22].
- Answer:
At 6. Slow start switched to congestion avoidance after reaching ssthreshold.
At 17. TCP reduces congestion window to ssthreshold/2 after a 3 DUP ACK event.
At 23. TCP goes into reduces congestion window size=1 after packet loss event.
- Answer:
7th Round. 63 segments are sent in slow start phase which ends in 6th round. Another 31 segments are sent in 7th round including 66th segment.
- Answer:
The size of congestion windows is 34 segments (MSS) in 10th round.

Part (c)

Departments	Network Address	Net Mask	Usable IP range	Broadcast Address
CS	208.16.7.0	255.255.255.128	208.16.7.1 - 126	208.16.7.127
SE	208.16.7.128	255.255.255.192	208.16.7.129 - 190	208.16.7.191
AI	208.16.7.192	255.255.255.224	208.16.7.193 - 222	208.16.7.223
CY	208.16.7.224	255.255.255.240	208.16.7.225 - 238	208.16.7.239

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Part (d)

- i. Solution:
PC-B will send an ARP reply with its MAC address (CC-49-DE-D0-AB-7D).
- ii. Solution:
It will send an ARP request for the MAC address of the default gateway (R) i-e (E6-E9-00-17-BB-4B)
- iii. Solution:
The IP address of PC-C (222.222.222.222)
- iv. Solution: Source (1A-23-F9-CD-06-9B) and Destination(49-BD-D2-C7-56-2A)

Question # 4:

[30 Minutes]

[4 + 2 + 4 = 10 Marks]

Part (a)

Step	Node (N')	D(u), p(u)	D(v), p(v)	D(w), p(w)	D(x), p(x)	D(z), p(z)
0	Y	∞	4,Y	8,Y	7,Y	2,Y
1	YZ	∞	4,Y	8,Y	7,Y	
2	YZV	11,V		7,V	7,Y	
3	YZVX	11,V		7,V		
4	YZVXW	10,W				
5	YZVXWU					

Forwarding Table Node Y	
Destination	Outgoing Link
U	Y, V
V	Y, V
W	Y, V
X	Y, X
Z	Y, Z

Part (b)

- i. Solution: The initial distance vectors of router w are: (u, v, w, x, y, z) = (3, 3, 0, 4, 8, ∞).
- ii. Solution: When the algorithm converges, router y has distance vectors (u, v, w, x, y, z) = (10, 4, 7, 7, 0, 2)

Part (c)

- i. Solution: Data plane is based on SDN controlled switches and uses flow table to forward traffic.

ii)	Match	Action
	IP Src = 10.1.*.* ; IP Dst = 10.3.*.*	Forward(3)
iii)	Match	Action
	Ingress Port = 1 ; IP Src = 10.1.*.* ; IP Dst = 10.3.*.*	Forward(4)
iv)	Match	Action
	Ingress port = 2 ; IP Src = 10.1.*.* ; IP Dst = 10.3.0.1 Ingress port = 2 ; IP Src = 10.1.*.* ; IP Dst = 10.3.0.2	Forward(3) Forward(4)

BEST OF LUCK ☺